

Machine Learning for Education

Dr Jakub Kuzilek

- Full names in Zoom
- Cameras on
- Microphones off
- Questions
 - raise hand (immediate response)
 - write in chat (at the end of the presentation)
- Missing in mandatory lecture → apology at least 24 hours in advance
- Problem → ask, write email, or contact me via Moodle

Learning objectives

- Know the basic concepts of machine learning
- Learn essentials of  programming language
- Understand how to process and analyse (educational) data
- Gain knowledge of essentials of educational data mining and learning analytics

Course structure

Order	Date	Topic
1	15. 04. 2025	Introduction
2	22. 04. 2025	Introduction to Machine Learning in Education; Introduction to R
3	29. 04. 2025	Vaclav Bayer, MSc (The Open University): Predictive Learning Analytics in practice (case of OUAlyse)
4	06. 05. 2025 14:00	Machine Learning - tasks, model training, data, evaluation; Data manipulation in R - Tidyverse
5	13. 05. 2025	Dr. Clara Schumacher (University of Potsdam): Pedagogical perspectives on Learning Analytics
6	20. 05. 2025	Unsupervised methods; Clustering (kmeans)
7	27. 05. 2025 14:00	Associate Professor Carrie Demmans Epp, PhD (University of Alberta): Language Models and Predicting Human Reading Performance
8	03. 06. 2025	dies academicus
9	10. 06. 2025	Dr. Milos Kravcik (DFKI): Scalable mentoring support with LLM chatbot & AI-powered psychomotor training
10	17. 06. 2025	Supervised methods; Caret & k-Nearest Neighbours
11	24. 06. 2025	Dr. Niels Seidel (CATALPA, FernUniversität in Hagen): Adaptive Learning
12	01. 07. 2025	Supervised methods; Naïve Bayes & Perceptron
13	08. 07. 2025	Supervised methods; Classification and Regression Tree, Random Forest
14	15. 07. 2025	Essay writing

Requirements

- ① Active participation in the external lectures
- ② Submission of all programming assignments
- ③ Essay

Programming assignments

- code is working (30 points)
- code is using the best practices, is readable, and well commented (30 points)
- code is producing correct results (40 points)
- deadline: 2 weeks after the lecture

PASS: $> 50\%$ of points from all assignments

- write an essay on selected topic
- the topic will be available after the 5th lecture
- each topic can be chosen by limited number of students
- proper referencing
- arguments needs to be supported by literature
- use of genAI is allowed, but mistakes are yours
- 1000 - 1500 words
- deadline: 15th July 2025

- we will be using Moodle for seminars, tasks and communication
- there will be an interactive component for each seminar
- Informed Consent Management system

Moodle

<https://moodlelab.hu-berlin.de/course/view.php?id=89>

Key: ML4ED25

Zoom

<https://hu-berlin.zoom-x.de/j/64124702126?pwd=j2H6Tys3Ryn6S9iMldrLpHiRjyCUjk.1>

Contact

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